SunAid: Smart Solar Monitoring and Control

A solar panel that reports its own voltage and switches loads on and off from your phone.

Built for **Sustainathon 26** by Team **Tech Innovators** (Problem statement: Nexus).

The problem

Small solar setups run blind. People can't see how much power the panel is producing or whether the load is using it well. SunAid reads the panel voltage with an ESP32, sends it live to a mobile app over WiFi, and lets the user switch a load with a relay, automatically or by hand.

Features

- Real-time solar voltage monitoring
- Automatic load switching when the panel voltage is high enough
- Manual override from the phone
- Device online/offline status
- Low cost, compact and easy to expand

Components

Component	Purpose
ESP32 Dev Kit	Brain of the system, WiFi
Voltage sensor module	Reads the solar panel voltage
1-channel relay module	Switches the load
6V solar panel	Power source being monitored
LED	Demo load

Component	Purpose
Breadboard and jumper wires	Connections

Wiring

ESP32 and breadboard (LED)

- D23 (yellow) to relay IN
- GND to ground rail
- 3V3 to positive rail

ESP32 and voltage sensor

- D35 (yellow) to sensor **S**
- Sensor **+** to positive rail
- Sensor **-** to ground rail
- Sensor VCC (red) to solar panel (+)
- Sensor GND (black) to solar panel (-)

Relay module and breadboard

- VCC to positive rail
- GND to ground rail
- IN to ESP32 D23 (the LED switches through the relay contacts)

Folder structure

```
firmware/sunaid_esp32.ino   ESP32 code (Arduino IDE)
app/App.js                  Mobile app (React Native + Expo)
```

Setup: ESP32

1. Install Arduino IDE and the ESP32 board package, then install the **Blynk** library.
2. Open `firmware/sunaid_esp32.ino`.
3. Fill in your Blynk template ID, auth token and WiFi name/password.
4. Select your ESP32 board and upload.

Blynk datastreams: **V0** = voltage (read), **V1** = load switch (write; 0 = allowed, 1 = off).

Setup: Mobile app

1. Install Node.js and the **Expo Go** app on your phone.
2. Run:

   ```
   npx create-expo-app sunaid-app
   cd sunaid-app
   ```

3. Replace the project's `App.js` with `app/App.js`.
4. Set `BLYNK_TOKEN` and `BLYNK_SERVER` at the top of `App.js`.
5. Run `npx expo start` and scan the QR code with Expo Go (same WiFi as the laptop).

Security note

Never commit your real Blynk token or WiFi password. Use placeholders in the repo.

Future work

- Voltage history graphs
- Energy (kWh) tracking
- Battery monitoring
- Notifications for low sunlight

Team

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